

Robust Regression Engine

Part A – Conceptual Foundation (Theory)

Question & Answer Notes | Advanced Supervised Learning Regression

1. What is Regularization in Machine Learning? Why is it needed?

Answer: Regularization is a technique used to reduce model complexity by adding a penalty to the loss function. It helps control overfitting, improves generalization, and makes predictions more stable on unseen data.

2. Difference between Ridge Regression (L2) and Lasso Regression (L1).

Answer: Ridge Regression uses an L2 penalty, which shrinks coefficients toward zero but generally keeps all features. Lasso Regression uses an L1 penalty, which can shrink some coefficients exactly to zero and therefore can perform feature selection.

3. What is Cross-Validation and why is it important?

Answer: Cross-validation evaluates a model across multiple train-validation splits instead of relying on one split. It gives a more reliable estimate of model performance and helps select models and hyperparameters that generalize better.

4. Explain the following cross-validation techniques:

K-Fold: The data is divided into K folds. The model trains on K-1 folds and validates on the remaining fold, repeating until every fold has been used for validation.

Stratified K-Fold: It maintains a similar target distribution across folds. For regression, target values can first be binned and then stratified.

LOOCV: One observation is used for validation while all remaining observations are used for training. This is repeated for every observation.

Time Series Split: Earlier observations are used for training and later observations for validation, respecting the time order.

5. Why are tree-based models less sensitive to feature scaling?

Answer: Tree-based models make decisions using feature thresholds and splits rather than distances or coefficient magnitudes. Therefore, changing the numerical scale of a feature generally does not change its ordering or the resulting tree splits.

Quick Revision

Topic	Main Purpose
Regularization	Control model complexity and overfitting
Ridge (L2)	Shrink coefficients
Lasso (L1)	Shrink coefficients and enable feature selection

Cross-Validation	Estimate generalization performance
Tree-Based Models	Learn threshold-based non-linear relationships

Based on Part A of the Robust Regression Engine assignment. Answers are kept brief as requested by the assignment.